CS331: Data Communications and Computer Networks

Switch and Router Network Design in Cisco Packet Tracer

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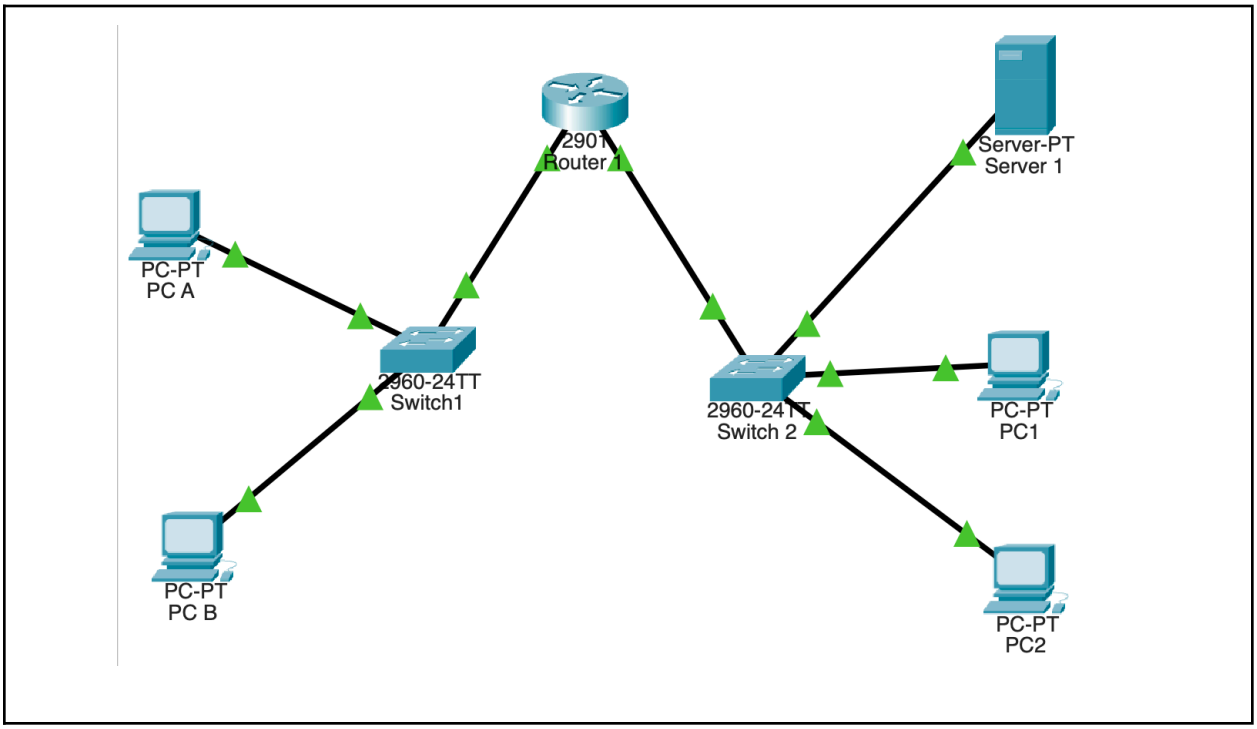
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Step 1: Topology

The network was built in Cisco Packet Tracer using one Cisco 2901 router (R1), two Cisco 2960 switches, four PCs and one server. Copper straight-through cables were used for all connections.

- Network A: PC A and PC B connected to Switch1, which connects to R1 GigabitEthernet0/0.
- Network B: PC1, PC2 and Server 1 connected to Switch 2, which connects to R1 GigabitEthernet0/1.



Step 2: Configure the Switch

No configuration is required on the switches. They forward frames at Layer 2 by default.

Step 3: Configure the Router

A. Network Design (Subnetting)

- Given block: 199.19.30.0/24
- Required subnet mask: /26 = 255.255.255.192

- Borrowed bits = 26 − 24 = 2 → number of subnets = $2^2 = 4$
- Host bits = 32 − 26 = 6 → addresses per subnet = $2^6 = 64$
- Usable hosts per subnet = 64 − 2 = 62
- Block size = 256 − 192 = 64, so subnets start at .0, .64, .128 and .192

Subnet	Network Address	Usable Range	Broadcast	Use
1	199.19.30.0/26	199.19.30.1 – .62	199.19.30.63	Network A (R1 G0/0)
2	199.19.30.64/26	199.19.30.65 – .126	199.19.30.127	Network B (R1 G0/1)
3	199.19.30.128/26	199.19.30.129 – .190	199.19.30.191	Future use
4	199.19.30.192/26	199.19.30.193 – .254	199.19.30.255	Future use

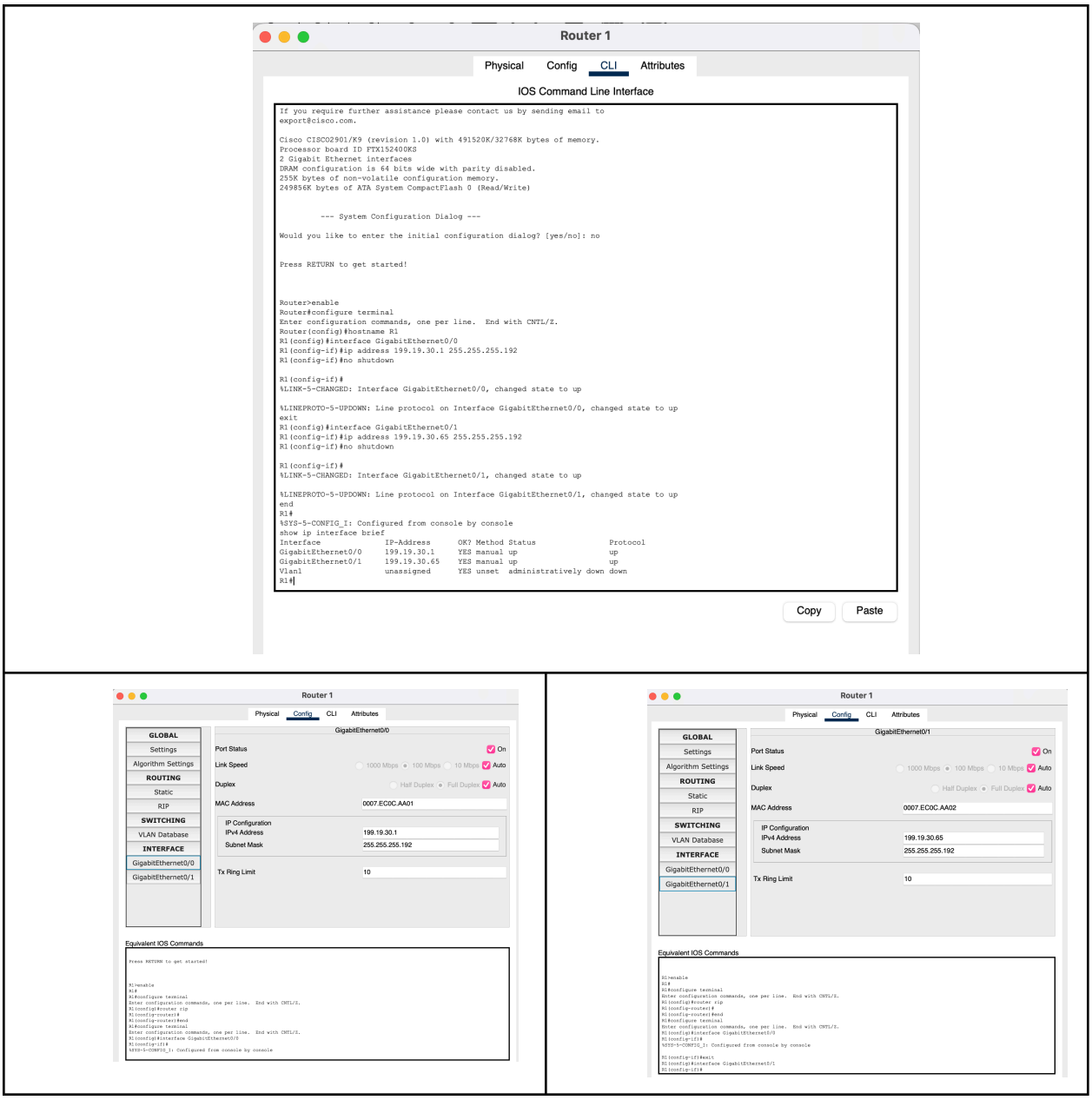
The first usable address of each subnet was assigned to the router interface, which acts as the default gateway for that subnet.

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	199.19.30.1	255.255.255.192	N/A
R1	G0/1	199.19.30.65	255.255.255.192	N/A
PC A	NIC	199.19.30.2	255.255.255.192	199.19.30.1
PC B	NIC	199.19.30.3	255.255.255.192	199.19.30.1
PC 1	NIC	199.19.30.66	255.255.255.192	199.19.30.65
PC 2	NIC	199.19.30.67	255.255.255.192	199.19.30.65
Web Server	NIC	199.19.30.68	255.255.255.192	199.19.30.65

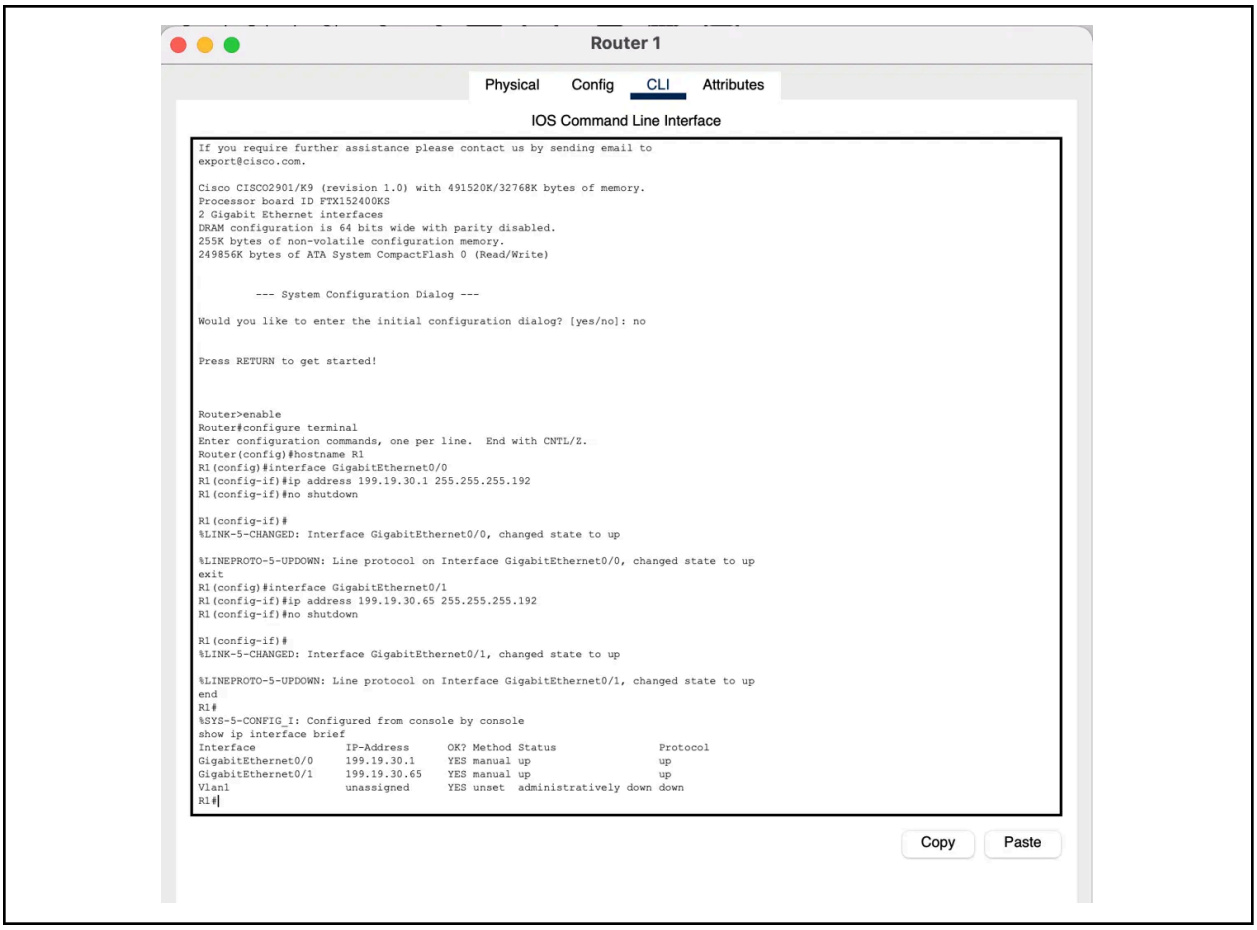
B. Router Interface Configuration

The router interfaces were configured through the CLI with the IP addresses from the table above.



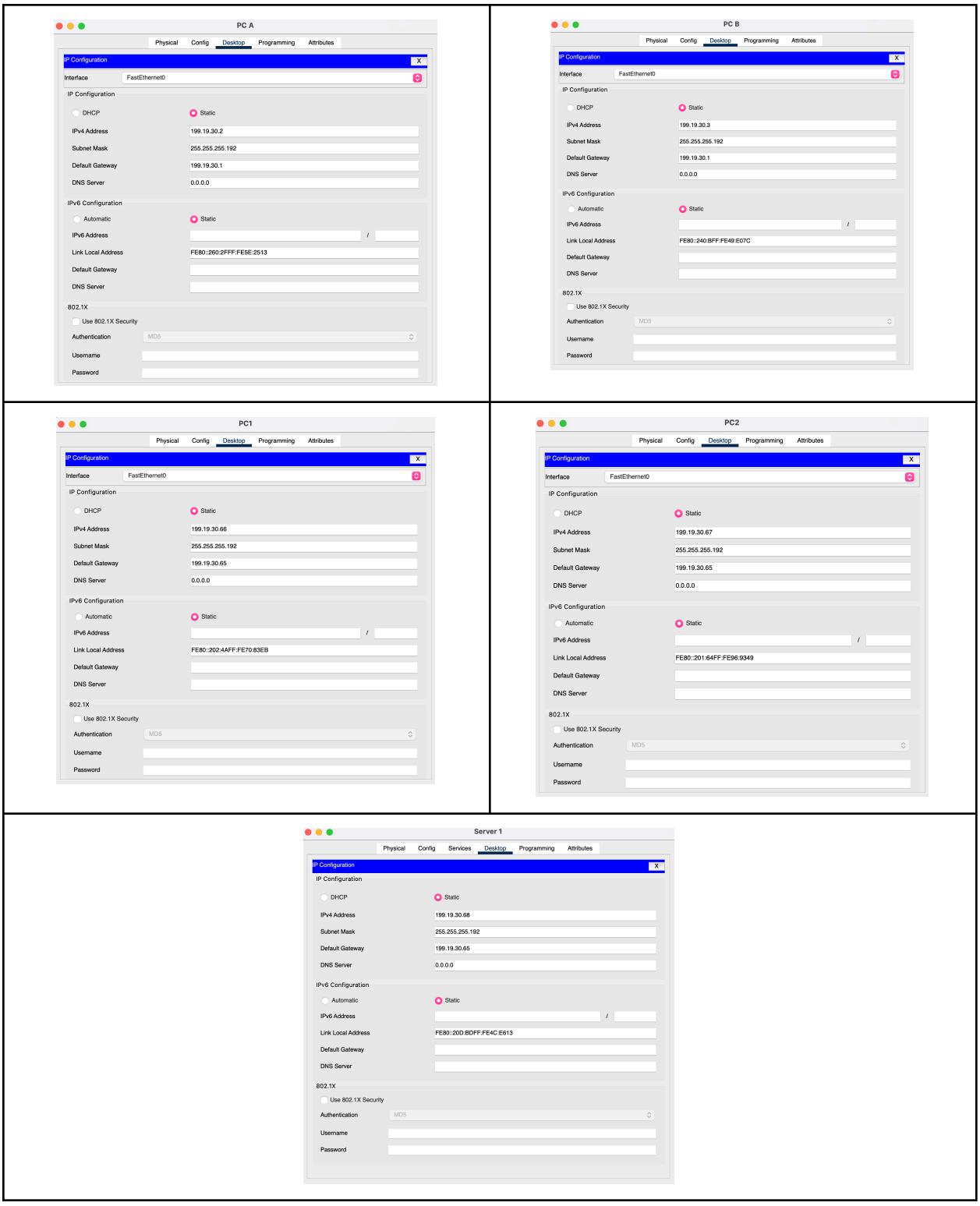
C. Activating the Interfaces

The `no shutdown` command was used to activate both interfaces. The `show ip interface brief` output confirms that GigabitEthernet0/0 and GigabitEthernet0/1 are both up/up.

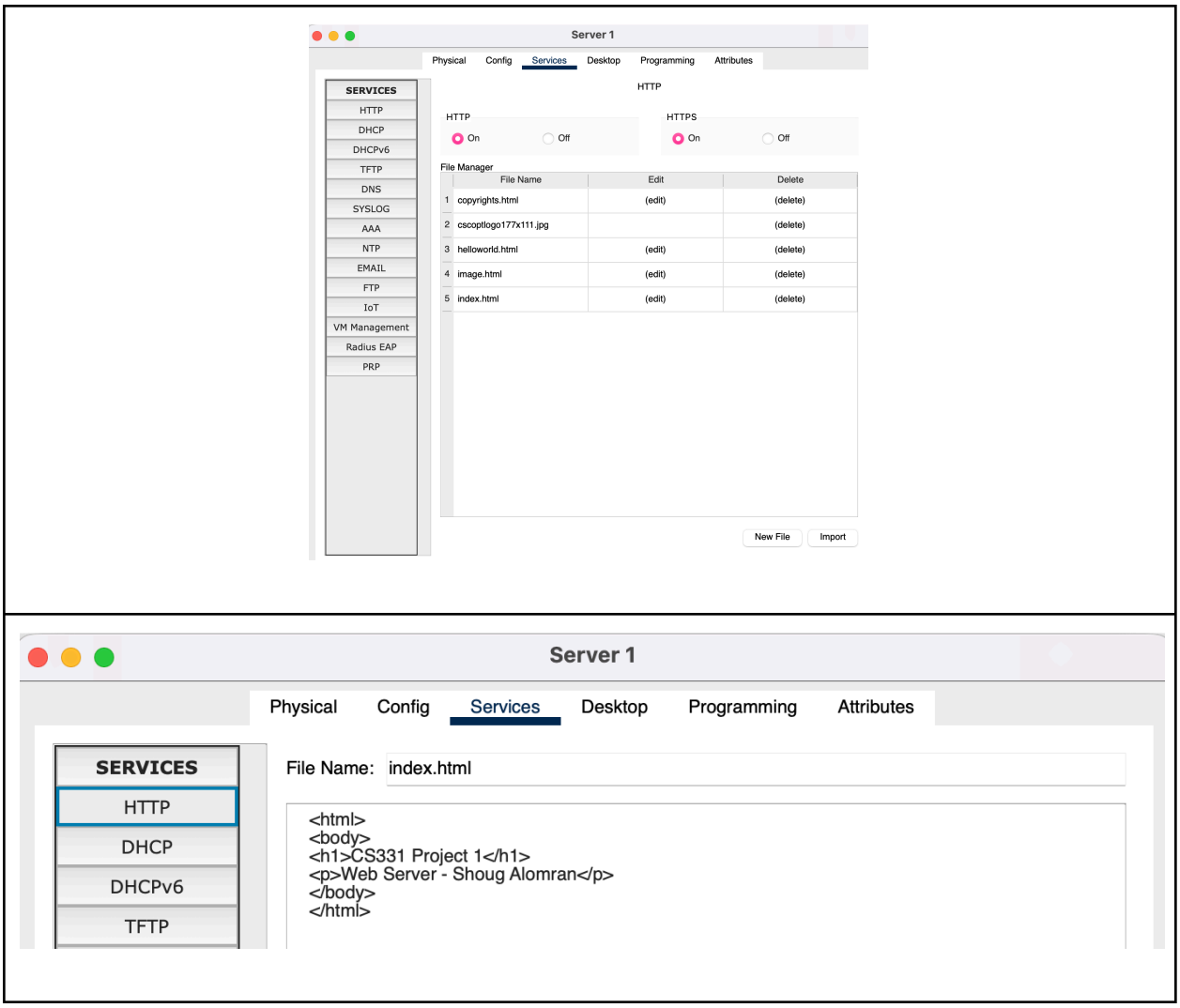


Step 4: Configuration on PCs and Web Server

Each end device was given a static IP address, subnet mask and default gateway from the addressing table. Devices in Network A (PC A, PC B) use gateway 199.19.30.1 (R1 G0/0), and devices in Network B (PC 1, PC 2, Web Server) use gateway 199.19.30.65 (R1 G0/1).



The server was configured as a web server: the HTTP service was turned on and index.html was edited to display a custom page.



Step 5: Verifying the Network

a. Ping from PC A to PC 2

PC A (Network A) successfully pinged PC 2 (199.19.30.67) on Network B, which confirms that R1 routes traffic between the two subnets. The TTL of 127 shows the packet passed through one router.

```
C:\>ping 199.19.30.67

Pinging 199.19.30.67 with 32 bytes of data:

Reply from 199.19.30.67: bytes=32 time=32ms TTL=127
Reply from 199.19.30.67: bytes=32 time=11ms TTL=127
Reply from 199.19.30.67: bytes=32 time<1ms TTL=127
Reply from 199.19.30.67: bytes=32 time<1ms TTL=127

Ping statistics for 199.19.30.67:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 32ms, Average = 10ms
```

b. PC B Accessing the Web Server

PC B on Network A opened http://199.19.30.68 in its web browser and loaded the web page hosted on Server 1 in Network B.

